Charles Backman

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Student founder building mobile apps, AI tools, and research software. Experience with local inference, browser-agent interfaces, source-grounded data products, and reproducible evaluation. Use AI coding tools while owning product direction, reviewing changes, and testing the result.

TECHNICAL SKILLS

Languages: Python, TypeScript, Swift, Java. **Application development:** React, React Native, Next.js, Django, FastAPI, SQL. **Research and quality:** NumPy, scikit-learn, PyTorch, Playwright, Git, GitHub Actions.

SELECTED ENGINEERING WORK

BlockScore | Founder & Product Lead

- Lead a five-person team building a native SwiftUI app for Canadian address research; direct product strategy, data sourcing, partnerships, and pre-launch execution.
- Develop source-labelled reports, property comparisons, and report-grounded questions with explicit coverage and unavailable-data states. In development; source code is private.

Evidence Board | Browser-agent research workspace

- Built a React/TypeScript workspace connecting claims to sources, with ten native WebMCP tools for inspection and proposed edits. People review changes; agents cannot approve their own proposals.
- Implemented revision checks, selective acceptance, undo, owner-scoped persistence, and browser tests. Public editable sample uses illustrative data.

Context | Offline relationship-memory app

- Built an Expo/React Native app with native iOS bridges, reviewed contact imports, QR sharing, validated backups, and optional on-device inference.
- Keep relationship data local and generated records editable; deterministic assistance supports the workflow when the model is unavailable.

AgriStress | Co-Founder & Lead Developer

- Work with University of Manitoba Plant Sciences researchers to incorporate longitudinal wheat and canola drone imagery into a multispectral segmentation platform.
- Develop provenance-aware manifests, field-aware evaluation, uncertainty maps, and GeoJSON review regions. Research prototype; external field validation remains incomplete.

Project Hope | Charity operations software

- Built Django/React workflows and an Electron desktop preview for contacts, volunteers, programs, and documents, with reviewed migration, organization permissions, and local AI adapters.
- Core operations work with AI disabled. The desktop sample uses fictional data; shared team use requires a separately hosted workspace.

Additional research: UFC Predictor recorded 63.55% accuracy on 1,709 chronological holdout fights. Quant Systems Lab replays 301,587 order-book messages with independent accounting and retained negative results. Historical studies, not live trading performance.

LEADERSHIP AND BUSINESS

Lead an AI club teaching students and teachers to build software. Founded Gear Garage (2024-2025), growing inventory to approximately 300 uniform items and an audience of 1,000+ followers. Co-organize a chess club teaching 40+ students; compete in national and provincial British Parliamentary debate.

EDUCATION AND RECOGNITION

St. John's-Ravenscourt, Winnipeg: Grade 12 honours student; expected graduation June 2027.
Certification: IBM SkillsBuild Artificial Intelligence Fundamentals.
Recognition: Second place, RRC Polytech business competition.